Cross-Reference Analysis: Battery Recycling Equipment Research

Consolidated Findings from Multiple Sources

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1. Executive Overview

This document synthesizes findings from three key sources:

1. **Executive Strategic Recommendation** - Cost-optimized hybrid approach (Chinese + Western equipment)
2. **PhD Research Buyer's Guide** - CAPEX-first analysis for 1-5 t/h scale
3. **Equipment Recommendations** - Detailed technical specifications

The PhD research document provides critical additional insights that refine and expand upon the previous recommendations.

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2. Key New Insights from PhD Research

2.1 ANDRITZ/Duesenfeld: The Game-Changer

Aspect	Previous Documents	PhD Research Update
Process	Standard mechanical recycling	Low-temperature vacuum (<80°C, ≤300 hPa)
HF Gas	Requires scrubbing	**Eliminates HF formation** - no scrubbing needed
Equipment	Requires stainless steel	Standard carbon steel sufficient
Electrolyte	Destroyed in drying	**Recovered at >99.5% purity** - revenue stream
CAPEX	40-60% savings vs BHS	**50% savings** vs conventional Western plants
Recycling Rate	72% (mechanical)	**91% at cell level**

Critical Finding: The Duesenfeld process fundamentally changes the economics by eliminating three major cost centers: HF gas scrubbing, stainless steel equipment, and continuous nitrogen consumption.

2.2 Additional Western Suppliers Identified

Supplier	Scale	Key Feature	Status
Erdwich (Germany)	50 kg - 5 t/h	Wet + dry shredding options	Targets exact 1-5 t/h range
ELDAN (Denmark)	4 t/h	ELDAN TOWER integrated system	Recently launched
MTB Recycling (France)	1-4 t/h	Containerized Drycell Box	Pilot running since Sept 2023
Primobius/SMS (Germany)	0.3 t/h	Mercedes reference, wet shredding	Lower throughput

2.3 Chinese Suppliers: Enhanced Analysis

Supplier	CAPEX	Verified Installations	Key Risk
Genox	500K– 1.2M	Partnerships with Sicon (Germany), Genox USA/UK	No ATEX certification
Xingmao	300K– 800K	**1.5 t/h USA, 1 t/h Singapore, 2 t/h South Korea**	After-sales logistics
Henan Doing	70K– 350K	Transparent pricing	Lowest tier quality

PhD Assessment: Xingmao provides **strongest evidence of international viability** with verified installations in USA, Singapore, and South Korea.

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3. Process Flow: Modernized Recommendations

Traditional 10-Step Process (Outdated)

1. Discharge → 2. Pre-sort → 3. Primary Shred → 4. Secondary Crush → 5. Fine Milling → 6. Screening → 7. Magnetic Sep. → 8. Eddy Current → 9. Air Classification → 10. Collection

PhD-Recommended 6-Step Process (CAPEX-Optimized)

1. **Deep electrical discharge** with energy recovery
2. **Integrated shredding tower** under N_2 (eliminates primary + secondary as separate steps)
3. **Vacuum drying** at <80°C with electrolyte recovery (Duesenfeld-type)

4. **Screening** at ~1mm cut point
5. **Separation train**: magnetic → eddy current → air classification
6. **Black mass collection** (cyclone + baghouse)

Key Changes:
- Integrated shredding towers (ANDRITZ, ELDAN, BCA) achieve ⩽15mm in single unit
- Fine milling often **eliminated** if shredding achieves adequate liberation
- Drying **before** separation improves downstream efficiency

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4. CAPEX Comparison: All Sources

Source	Capacity	CAPEX	Notes
BHS-Sonthofen	1-15 kt/yr	€8-20M ($9-22M)	3+ proven plants, >97% recovery
ANDRITZ/ Duesenfeld	2 t/h	Est. €5-12M	50% less than BHS
Western Alternatives	1-5 t/h	€3-8M	Erdwich, ELDAN, MTB
Chinese (Genox)	1-3 t/h	500K− 1.2M	European partnerships
Chinese (Xingmao)	1-2 t/h	300K− 800K	Verified international installs
Chinese (Henan Doing)	500-3000 kg/h	70K− 350K	Budget option

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5. Safety Systems: Critical Insights

PhD Research Key Findings

System	PhD Assessment	Implication
ATEX Certification	**Critical gap** for Chinese equipment	Black mass dust = Zone 20/21/22
CE Marking	Often self-declared, variable rigor	Third-party assessment rare
Wet Shredding	Eliminates inert gas needs	Reduces CAPEX significantly
Cryogenic	High OPEX, not recommended	Liquid nitrogen ongoing cost

Recommended Safety Architecture (Hybrid)

Component	Recommended Source	Rationale
Shredding (critical)	Western: BHS, ANDRITZ, Erdwich	ATEX compliance, insurance acceptance
N_2 Systems	Western: Generon	Proven reliability
Gas Detection	Western: Drager/Honeywell	ATEX certified
Fire Suppression	Western: Siemens/Tyco	Insurance requirement
Separation Equipment	**Either Western or Chinese**	Lower safety stakes

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6. Black Mass Quality: Realistic Expectations

PhD Research Findings on Purity Limits

Impurity	Typical Level	GB/T 45203-2024 Grade 1 Target
Copper	1-10%	<0.5%
Aluminum	1-5%	<1%
Iron	0.5-3%	<1%

Key Insight: Meeting Grade 1 standards (Ni+Co>35%, Li>3.5%) is:
- **Achievable** with production scrap (consistent chemistry)
- **Extremely challenging** with mixed end-of-life batteries

Feedstock Chemistry Impact

Chemistry	Black Mass Value	Processing Challenge
NMC	Highest (Ni 9-25%, Co 5-16%, Li 2-4%)	Standard processing
LFP	Lithium only	Iron creates separation issues
Mixed	Requires sorting	Must segregate NMC from LFP

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7. Consolidated Recommendations

By Regulatory Environment

Jurisdiction	Recommended Approach	Estimated CAPEX
EU/ATEX-regulated	ANDRITZ/Duesenfeld pathway	€5-12M
Outside ATEX (SE Asia, Middle East, Africa)	Chinese from Genox/Xingmao	500K− 1.5M
Aggressive CAPEX minimization	Wet shredding + Chinese separation	300K− 800K

Universal Recommendations

1. **Invest disproportionately in screening** - Multi-deck precision screens have highest marginal impact on purity
2. **Upstream chemistry sorting** - XRF-based sorting separates NMC from LFP before processing
3. **Don't eliminate fine milling entirely** - Only if shredding tower achieves adequate liberation
4. **Drying before separation** - PhD research confirms this improves downstream efficiency

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8. ROI Analysis: Updated

Consolidated from All Sources

Scenario	Throughput	Annual Output	Revenue	Payback
Conservative	500 kg/hr	300 tons	$900K	36+ months
Expected	**1,000 kg/hr**	**810-1,500 tons**	**$3-6M**	**24-30 months**
Optimistic	2,000 kg/hr	3,000+ tons	$12M+	18 months

Critical Success Factors (All Sources Agree):
1. Feedstock security (2-3 year agreements)
2. Black mass offtake contracts
3. Safety system investment (non-negotiable)
4. Process optimization expertise

9. Summary: Cross-Document Consensus

Factor	Executive Recommendation	PhD Research	Consensus
Best Western	BHS-Sonthofen	ANDRITZ/ Duesenfeld	Both credible; Duesenfeld more cost-effective
Best Chinese	GEP ECOTECH	Xingmao	Xingmao has best international verification
Hybrid Approach	Recommended	Also recommended	Strong consensus
Safety Systems	Western mandatory	Western mandatory	Absolute consensus
CAPEX Range	180K−350K| 300K-1.5M (Chinese) / €5-12M (Western)	Reasonable range	
Process Steps	10 stages	6 stages (modernized)	PhD provides optimization

10. Final Assessment

The PhD research document significantly enhances the strategic recommendation by:

1. **Validating the hybrid approach** as a sound strategy
2. **Adding ANDRITZ/Duesenfeld** as a compelling alternative to BHS
3. **Providing realistic CAPEX ranges** for different supplier tiers
4. **Identifying process flow optimizations** that reduce equipment count
5. **Clarifying impurity limits** and quality expectations
6. **Highlighting ATEX as a critical decision factor**

Bottom Line: For a CAPEX-first buyer at 1-5 t/h scale, the optimal path is:
- **EU/ATEX markets:** ANDRITZ/Duesenfeld (€5-12M)
- **Other markets:** Chinese from Xingmao or Genox (500K− 1.5M)
- **Both paths:** Maintain Western safety systems throughout